BIJOY BHASKAR LALL

New Delhi, India, 110057 | +91 9958502159 | bijoylall2012@gmail.com | https://www.linkedin.com/in/bijoy-lall/

EDUCATION

Birla Institute of Technology and Science (BITS) Pilani, Pilani, India **Nov 2020 – May 2024**
Secured a Bachelor of Engineering in Mechanical Engineering with *Distinction* **(CGPA 9.29/10)**
Merit Scholarship for being in the top 2% of the batch

PROFESSIONAL EXPERIENCE

Mercedes-Benz Research and Development India (MBRDI), Bengaluru | *CAE Analyst* **Oct 2024 – Present**

- Develop and improve physics-based AI models and digital twin workflows that combine engineering simulations with machine learning to model real-world system behaviour, optimize production workflows, and improve planning efficiency.
- Perform automotive interior component simulations using ANSA and ABAQUS across 2+ vehicle carlines, 10+ load cases, and 30+ seat hardware tests to support design evaluation and development.
- Developed Python-based ML and data analytics frameworks to predict occupant comfort from objective metrics, correlating occupant feedback and pressure, thus reducing reliance on physical testing and enabling data-driven comfort assessment.
- Developed Python automation workflows that reduced analysis time from 30 days to 5 days and saved 2 days per project.
- Co-inventor on 3 patents (patent pending) to prevent serious injuries in vehicle crash scenarios.

Mercedes-Benz Research and Development India (MBRDI), Bengaluru | *Student Intern* **Jul 2023 – Dec 2023**

- Initiated a study on the influence of seat design on comfort using data analytics; built and analysed a database of design parameters and comfort metrics using Python, ABAQUS, and machine learning techniques.
- Developed an efficiency improvement process projected to save €70,000 annually and coordinated weekly managerial meetings for 80+ colleagues covering technical and leadership updates.

Birla Institute of Technology and Science Pilani, Pilani, India | *Teaching Assistant – Theory of Machines* **Jan 2023 – May 2023**

- Taught classes of 200+ students on using AutoCAD software to solve kinematic analysis problems involving link-chain mechanisms; conducted doubt-solving sessions.
- Assisted in creation of class test questions as well as in invigilation and paper correction.

Indira Gandhi Centre for Atomic Research (IGCAR), Kalpakkam, India | *Research Intern* **Jun 2022 – Jul 2022**

- Developed a mathematical model of the robot manipulator arm at IGCAR using the Denavit-Hartenberg approach; analysed the kinematics of the manipulator arm using MSC ADAMS and AutoCAD.
- Authored and presented a research report to a panel of distinguished scientists at IGCAR.

ACADEMIC PROJECTS

Kinematics of Humanoids for Bipedal Locomotion

- Developed a mathematical model of a bipedal humanoid robot using the Denavit-Hartenberg approach.
- Validated the kinematics of the model using MATLAB to conduct complex matrix transformation calculations.
- Used MSC ADAMS to digitally simulate and validate the model over its degrees of freedom.

Vibrational Analysis of Ear Ossicles

- Established a correlation between the resonance frequency of the ear structure and changes in its density due to disease.
- Modelled the ear structure using Fusion 360, performed modal analysis in ANSYS across different material properties, and used exponential curve fitting to quantify the relationship.

Exergy Analysis of Low Temperature Combustion (LTC) Engines

- Developed a model of a two-zone LTC engine using MATLAB to model burned/unburned zones, temperature variation, combustion dynamics, and multiple chemical species using NASA-polynomial thermodynamic properties.
- Applied the first and second law of thermodynamics to evaluate energy transfer and exergy destruction from combustion, heat transfer, and friction; validated the model through pressure-volume and pressure-crank angle profiles.

TECHNICAL SKILLS & CERTIFICATIONS

- **CAE, CAD & Simulation:** ANSA, ABAQUS, ANSYS, LS-DYNA, HyperView, METAPOST, MSC ADAMS, AutoCAD, Fusion 360
- **Programming & Data Analytics:** Python, MATLAB, Excel for Data Analytics
- **Certifications:** AI Foundations: Machine Learning, Python for Data Science and Machine Learning

EXTRA CURRICULAR ACTIVITIES

- Topic Driver, Innovation Cluster, MBRDI (May 2025 - Present): Lead ideation and filing for 10+ patents.
- Member of the BITS Pilani Cricket Team (March 2022 - May 2023): Organized annual inter-college tournaments and coordinated the preparation of facilities. School team captain; won accolades at multiple inter-school tournaments.
- Founder & Vice-President, Model United Nations (MUN) Club, Modern School (2019 - 2020): Won national-level accolades.
- I.P. Gupta Scholarship for Best Male Scholar (2019): Awarded to the top 1% of students in the science stream.
- Social Venture: Sardar Bhagwant Singh award for excellence in research work (2018): Outreach program to educate students on the constructive use of technology. Held classes and sessions at school and undertook door-to-door community outreach.
- Volunteer, WWF-India (2017 - 2018): Conducted wildlife workshops and contributed research for WWF-India publications.